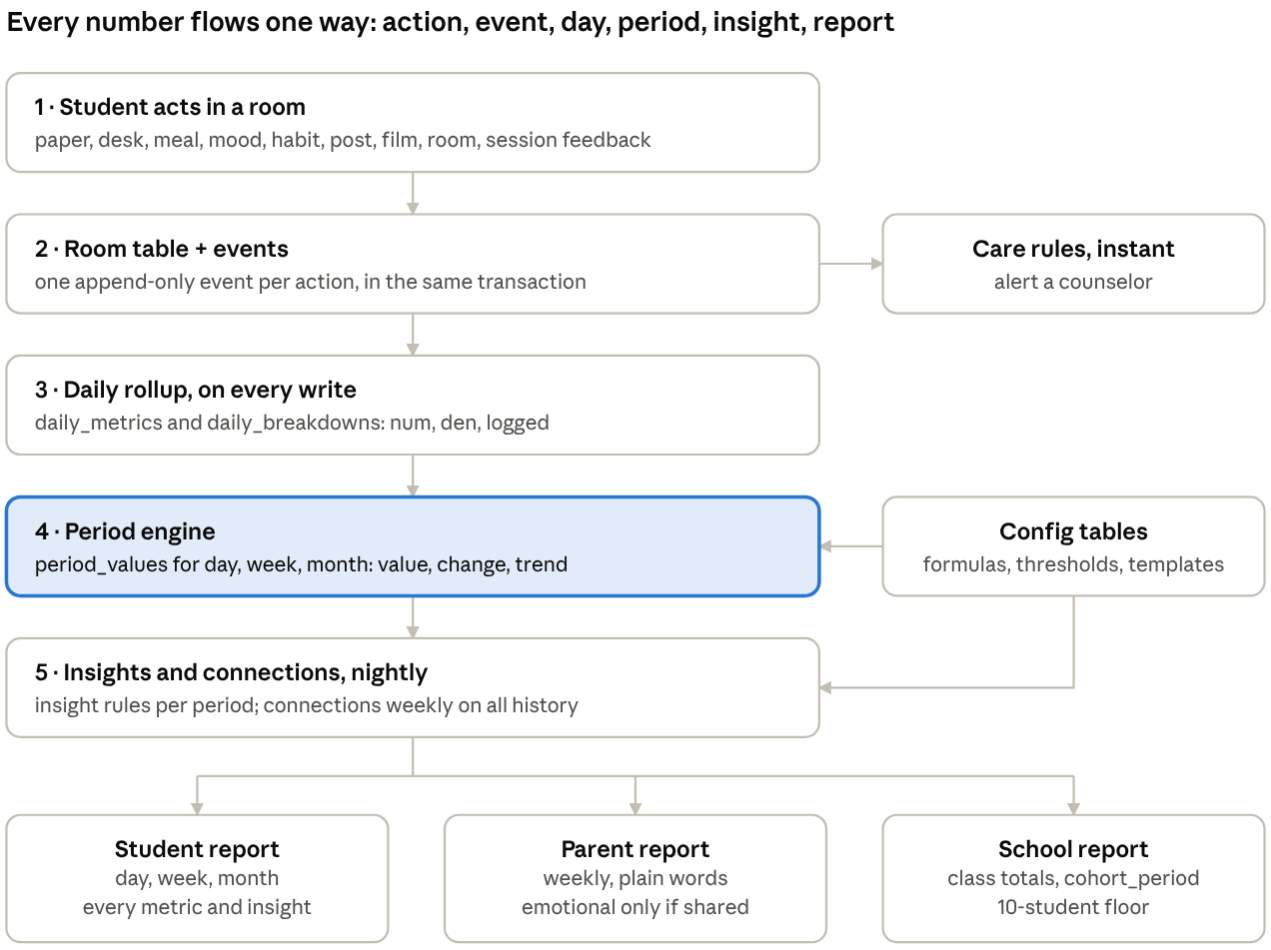
DHI Master Spec: Metrics, Storage, Insights and Build Roadmap v1

Oct 13, 2026 · @Dhi

This is the single build spec for the Growth report: every metric, how it is calculated, where it is stored, which screen uses it, and the order to build it all from today's demo repo. It extends the architecture doc and the metrics doc; where they differ, this doc wins.

Pipeline at a glance

Every number on every report passes through the same five steps, and nothing ever flows backwards.



metric pipeline · 5 steps, 3 reports, care branch

One action becomes one event; events roll up into daily values as they arrive; any day, week or month is then a sum over days; insight rules read those period values; the three reports read only the results. Care rules sit on the side and act within seconds.

Storage

The metric layer adds eight tables and two config tables beside the shared `events` log from the architecture doc. Raw room tables stay as they are; every number on a report is rebuilt from `events`, so nothing is ever lost by changing a formula.

Table	One row per	Key columns	Written by	Read by
`events`	room action	event_id, student_id, institute_id, class_id, room, type, logged_for, occurred_at, created_at, topic_id, subject_id, payload jsonb, schema_ver	each room's server function, same transaction as its own table	daily rollup
`session_feedback`	finished study session	session_id, student_id, source (desk, room), subject_id, started_at, minutes, feel (energised, okay, drained), focus 1–5	feedback card	daily rollup
`daily_metrics`	student × metric × day	value, num, den, logged bool, backfilled bool	daily rollup	period engine

Table	One row per	Key columns	Written by	Read by
`daily_breakdowns`	student × metric × dimension × key × day	dim (subject, topic, mode, mood_family, weekday, time_band, length_band, food_group, habit, context, tag, mistake_type), key, num, den	daily rollup	period engine, insight rules
`period_values`	student × metric × period	period_type (day, week, month), period_start, value, num, den, logged_days, prev_value, start_value, change_prev, change_start, trend	period engine	reports, insight rules
`insights`	student × period × rule hit	rule_id, room, profile, priority, params jsonb, audience (student, parent, school)	insight engine	reports
`connections`	student × pre-registered pair	pair_id, n_days_a, n_days_b, mean_a, mean_b, ci_low, ci_high, q, shown bool, feedback	connections engine	student and parent reports
`cohort_period`	class × metric × period (× breakdown key)	n_students, num, den, value, distribution jsonb, suppressed bool	cohort rollup	school report
`sharing_settings`	student	parent_level (none, summary), set_at	student settings	parent report

Table	One row per	Key columns	Written by	Read by
`metric_catalog` (config)	metric	id, name, profile, unit, daily formula id, period aggregation, direction, clear-rise per period, audiences, min_logged_days	admin, versioned	every engine
`insight_rules` (config)	rule	id, room, profile, condition id, template keys per language, priority, audience	admin, versioned	insight engine

Rules that keep the numbers honest:

- `num` and `den` are stored separately, so a week's accuracy is Σnum ÷ Σden, never an average of daily percentages.
- A day with no log is stored with `logged = false`, never as 0.
- `period_values` for finished weeks and months are frozen. The current period is computed live and cached for 5 minutes.
- Changing a formula bumps the metric's version and rebuilds `daily_metrics` from `events`.

Metric catalog: Academic profile

Every metric has an ID that never changes; reports, insight rules and tests refer to the ID, not the name. Daily values are stored as numerator ÷ denominator; a period sums both. "S P Sch" = shown to student, parent, school.

ID	Metric	Daily value (num ÷ den), from	Period value	Clear rise, week / month	Breakdowns	Seen by
A01	Daily paper on time	papers submitted before due ÷ papers due that day · `paper_submitted`, `paper_missed`	days on time of days with a paper	+2 / +4 days	subject	S P Sch

ID	Metric	Daily value (num ÷ den), from	Period value	Clear rise, week / month	Breakdowns	Seen by
A02	Accuracy	MCQ items right ÷ items attempted · `item_answered`	Σright ÷ Σattempted, %	+10 / +8 pts	subject, topic	S P Sch
A03	Understanding in written answers	mean of teacher rubric (understanding, structure, expression, each 0–4) · `remark_given`	mean of all written answers, 0–4	+0.4 / +0.4	subject	S P Sch
A04	Recall after 7 days	review items right ÷ review items asked (items first seen ⩾ 7 days earlier) · `item_answered` with is_review	Σ ÷ Σ, %	+10 / +8 pts	subject, topic	S P Sch
A05	Old mistakes now fixed	past-wrong items answered right ÷ past-wrong items re-asked	Σ ÷ Σ, %	+10 / +8 pts	subject	S P Sch
A06	Hint reliance	items with a hint opened ÷ items attempted · `hint_opened`	Σ ÷ Σ, %; lower is better	−15 / −10 pts	subject, topic	S Sch
A07	Why answers were wrong	wrong items by type ÷ wrong items (rule in Calculation rules)	share per type	shown, no trend	mistake_type × subject	S P Sch
A08	Answer pace	seconds per MCQ	median, vs own median	shown only	subject	S
A09	Topics mastered	topics reaching Internalised that day	count in period; total to date	+1 / +3	subject	S P Sch
A10	Desk study days	1 if a sitting of 3+ turns	days	+2 / +4 days	none	S P
A11	Desk time by subject	active minutes per subject ÷ all active minutes (gap between messages, capped at 5 min) · `sitting_closed`	share per subject	shown, no trend	subject, topic	S P Sch
A12	Active learning share	sittings in mock, recall, write or paper mode ÷ sittings	Σ ÷ Σ, %	+15 / +10 pts	mode	S Sch
A13	Recall rounds right	recall answers right ÷ asked · `recall_answered`	Σ ÷ Σ, %	+10 / +8 pts	subject	S Sch

ID	Metric	Daily value (num ÷ den), from	Period value	Clear rise, week / month	Breakdowns	Seen by
A14	Weak-topic follow-up	topics studied on the desk within 3 days of a wrong paper answer ÷ topics got wrong	Σ ÷ Σ, %	+20 / +15 pts	subject	S Sch
A15	Modules finished	films kept · `module_kept`	count; total to date by category	+1 / +2	category, subject	S P Sch
A16	Module check accuracy	check answers right ÷ asked · `module_check_answered`	Σ ÷ Σ, %	shown	subject	S Sch
A17	Module effect on papers	A02 on the module's topic in 7 days after − 7 days before (4+ items each side)	points, per module	shown	topic	S P Sch
A18	Doubts resolved	doubts marked resolved ÷ doubts asked · `doubt_resolved`	Σ ÷ Σ, %	+15 / +10 pts	subject	S Sch
A19	Help given to classmates	answers marked helpful or mentor-verified	count	+1 / +2	subject	S P Sch
A20	Pre/post paper gain (trial)	day-21 matched paper % − day-1 paper %	points, once	shown	subject	S P Sch

Metric catalog: Emotional and expression profile

These metrics describe how the student feels and expresses themselves, so they carry the strictest audience rules: full detail to the student, a summary to parents only if the student shares it, and class totals of 10 or more to the school. None of them is ever rewarded with coins or milestones.

ID	Metric	Daily value, from	Period value	Clear rise, week / month	Breakdowns	Seen by
E01	Mood check-in days	1 if a mood was logged · `mood_logged`	days	+2 / +4	none	S P Sch(total)
E02	Stressed days	1 if mood is stressed or anxious	days; lower is better	−2 / −4	context, weekday	S P(level) Sch(total)
E03	Heavy days	1 if mood is sad or heavy	days; lower is better	−1 / −2	context, weekday	S, counselor via care rules

ID	Metric	Daily value, from	Period value	Clear rise, week / month	Breakdowns	Seen by
E04	Stress level	(E02 + E03 days) ÷ logged days	Low under 15%, Moderate 15–35%, High over 35%	word changes	none	S P(if shared)
E05	Mood swing	mean of |lightness today − lightness on the previous logged day| (lightness 1–5)	Low under 0.8, Medium 0.8–1.5, High over 1.5	−0.3 / −0.3	none	S
E06	Mood mix	days per family: light (best, happy, motivated, determined, content), steady (tired), heavy (stressed, anxious, sad, heavy)	days and share	shown	mood_family	S
E07	When stress shows up	stressed or heavy days by context: day before a paper, after 3+ study hours that day, other	share per context, last 28 days	shown	context	S, Sch(total)
E08	Note days	1 if the mood note has 10+ words	days	shown	none	S
E09	Felt okay or better after studying	sessions marked energised or okay ÷ sessions with feedback · `session_feedback`	Σ ÷ Σ, %	+15 / +10 pts	subject, time_band, length_band	S P(if shared) Sch(total)
E10	Focus rating	mean focus 1–5 of sessions	mean	+0.3 / +0.3	subject, time_band, length_band	S Sch(total)
E11	Writing days	1 if a post or a 50+ word draft · `post_published`	days	+1 / +2	none	S P
E12	Words per post	words in each post	median	shown	none	S P Sch
E13	Vocabulary range	moving type-token ratio over 50-word windows, per post	median; vs first 3 posts	+0.03	none	S Sch
E14	Structure	posts with paragraphs or a list ÷ posts	Σ ÷ Σ	shown	none	S
E15	What they write about	posts by tag ÷ posts	share per tag	shown	tag	S

ID	Metric	Daily value, from	Period value	Clear rise, week / month	Breakdowns	Seen by
E16	Peer engagement	comments written + replies received	counts	shown	none	S
E17	Asked for help	help requests	count, never a trend, never scored	none	none	S, counselor

Lightness scale: best and happy 5; motivated, determined and content 4; tired 3; stressed, anxious and sad 2; heavy 1. It is used only inside formulas and is never shown.

Metric catalog: Health and consistency profile

Health metrics measure rhythm and balance, never amounts: no calories, weight or portion size enters any formula.

ID	Metric	Daily value, from	Period value	Clear rise, week / month	Breakdowns	Seen by
H01	Breakfast before 11	1 if a breakfast plate eaten before 11:00 · `plate_logged`	days	+2 / +4	none	S P Sch
H02	Food groups	groups present that day of 5: grain, dal or protein, vegetable, dairy, fruit	median day	+1 / +1	food_group (days present)	S P Sch(total)
H03	Meal times	eaten_at per slot	median time and spread (MAD, minutes) per slot	spread −20 min	slot	S P
H04	Late dinners	1 if dinner after 22:00	days; lower is better	−2 / −4	none	S P
H05	Packaged or fried items	items tagged packaged or fried ÷ items	share; shown without judgement	shown	none	S
H06	Bedtime	bed_at · `sleep_logged`	median time; spread (MAD)	−20 min	weekday	S P Sch(total)
H07	Sleep length	wake_at − bed_at	median	shown	weekday	S P
H08	Habit days kept	1 if any habit ticked; also per habit · `habit_ticked`	days	+2 / +4	habit	S P

ID	Metric	Daily value, from	Period value	Clear rise, week / month	Breakdowns	Seen by
H09	Forgiving streak	consecutive days with any log, one miss per 7 allowed	current and longest	shown	none	S P
H10	Habit feels automatic	weekly 4-question self-rating, 1–7 · `automaticity_rated`	mean	+1	habit	S
H11	Late-night study room time	minutes present after 23:00 · `presence_span`	minutes; lower is better	−30 / −60 min	weekday	S P Sch(total)
H12	Study room days	1 if 20+ minutes present	days	+2 / +4	none	S P
C01	Days showing up	1 if any meaningful log in any room	days	+2 / +4	none	S P Sch
C02	Room coverage	1 per room with a log that day	days per room	shown	room	S Sch

New data each room must capture

Nine additions to today's rooms make every metric above computable; without them the report falls back to counts. Each is small, but the topic and subject tags are the foundation of every academic insight.

Room	Add	Why	Effort for the student
Exams	`topic_id`, `subject_id`, `difficulty` and server-only `answer_index` on every question; each wrong MCQ option tagged `misread` or `concept` by the paper author	A02, A04, A05, A07, topic heat map	none
Exams	teacher rubric (understanding, structure, expression, 0–4) on each written answer	A03	none; teacher taps 3 scores

Room	Add	Why	Effort for the student
Exams	review scheduler: 40% of each daily paper drawn from past-wrong items and topics seen 3, 7 and 21 days ago	A04, A05	none
DHI desk	subject and topic on each sitting: picked from a chip, or inferred from the paper or module it was opened from	A11, A14	one tap when not inferred
DHI desk, study room	session feedback card when a sitting of 10+ minutes or a room stay of 20+ minutes ends: feel (energised, okay, drained) and focus (1–5); skippable	E09, E10, connections	two taps
Modules	2 check questions after each academic film, tagged to its topic	A16, A17	30 seconds
Meals	`food_group` and neutral `processing` tag on every catalog item; custom dishes get a group when DHI fills them	H02, H05	none
Sleep (new)	bedtime and wake time, prefilled from yesterday	H06, H07	two taps
Habits	weekly automaticity question set (4 items)	H10	20 seconds a week

The blog, mood, habits and study room already capture what their metrics need.

Calculation rules

The same seven rules apply to every metric, so one engine computes all of them.

1. Day boundary. A day runs 04:00 to 04:00 IST. A meal at 01:00 or a mood logged at 00:30 belongs to the day before.

2. Period value. For a day, week (Mon–Sun), month (calendar) or custom range:

- share metrics: Σnum ÷ Σden over the period
- day-count metrics: number of days with value 1, shown as "5 of 7"
- time metrics: median of daily values; spread = median absolute deviation in minutes
- means (rubric, focus, automaticity): mean of all individual answers in the period, not of daily means
- `logged_days` = days in the period with `logged = true`; printed beside every value

3. Change. `change_prev` = this period − the previous period of the same length. `change_start` = this period − the student's first period of that length on DHI. Both are null if either side has no logs.

4. Trend word. d = change_prev × direction (+1 if higher is better, −1 if lower is better). Rising if d ⩾ the metric's clear-rise, Dipping if d ⩽ −clear-rise, else Steady. The day view shows no trend word.

5. Breakdowns. Each breakdown key gets its own num and den in `daily_breakdowns`, so "accuracy in Physics this week" is Σnum ÷ Σden for key = Physics. A breakdown value is shown only with 5+ items behind it (for example, 5 Physics questions).

6. Mistake type (A07). Each wrong MCQ answer gets one type, in this order:

1. **Misread:** the chosen option is tagged `misread` by the paper author.
2. **Careless:** the student got this topic right on 2 of their last 3 attempts and answered in under half their own median time.
3. **Concept not clear:** everything else.

Written answers take the type the teacher picks in the remark, if any.

7. Session feedback. The card appears when a desk sitting of 10+ minutes or a study-room stay of 20+ minutes ends. Subject comes from the sitting. Time band is set by the start time: before 18:00, 18:00–21:00, after 21:00. Length band: under 40 minutes, 40 minutes or more. Skipped cards are stored as skipped and leave E09 and E10 unchanged.

Stress context (E07). "Day before a paper" = a paper is due the next day. "After 3+ study hours" = desk plus room minutes that day ⩾ 180. A day can count in only one context, in that order.

Sincerity. If more than 30% of a metric's entries in a period were logged 36+ hours late, the value shows "some entries added later" and earns no milestones.

Insight rules library

Every sentence in "What DHI noticed" and every profile headline is produced by a fixed rule with a fixed template, so the same data always gives the same words and each can be tested. Rules run after `period_values` are ready; each hit is stored in `insights` with its

parameters.

ID	Room	Fires when	Template (English)	Audience	Priority
IA1	Exams	best and worst subject on A02 differ by 15+ pts, each with 10+ items	"{best} is carrying your marks; {worst} is the gap."	S P Sch	1
IA2	Desk + Exams	the lowest-accuracy subject has a desk-time share (A11) below 1 ÷ number of subjects	"{subject} gets {share} of your desk time but has your lowest accuracy."	S P Sch	1
IA3	Exams	a topic wrong 3+ times in the period, with hints on 50%+ of its items	"{topic}: wrong {n} times, hint used on {h}. The idea has not landed yet."	S Sch	1
IA4	Exams	one mistake type is over 50% of wrong answers	"Most wrong answers are {type}, not {other}."	S P	2
IA5	Modules	A17 ≥ +5 pts for a module	"After '{module}', {topic} accuracy went from {before} to {after}."	S P	2
IA6	Desk	A12 below 40%	"Most sittings were explanations. Recall rounds build memory faster."	S	3
IA7	Exams	A05 up by the clear-rise amount	"You fixed {n} old mistakes this {period}."	S P	2
IE1	Mood	E02 changed by its clear-rise	"Stress came on {days}, {fewer or more} than last {period}."	S	1
IE2	Mood	60%+ of stressed or heavy days in the last 28 fall in one context, with 3+ such days	"Stress tends to show up {context}."	S P(if shared) Sch(total)	1
IE3	Sessions	a subject with 3+ sessions where 60%+ were "drained"	"{subject} sessions often leave you drained."	S P(if shared) Sch(total)	1
IE4	Sessions	focus differs by 0.8+ between time bands	"Your focus is {hi} before {time} and {lo} after it."	S	2
IE5	Sessions	E09 differs by 30+ pts between length bands	"Sessions under 40 minutes felt better."	S	3
IE6	Blog	E12 or E13 up vs the first 3 posts	"Your posts are longer and more varied than your first ones."	S P	3
IH1	Meals	one food group present on 3 or fewer days of 7	"{group} is the group most often missing."	S P	2

ID	Room	Fires when	Template (English)	Audience	Priority
IH2	Sleep	H06 changed by 20+ min	"Bedtime {n} min {earlier or later}."	S P	1
IH3	Meals	H04 dropped by 2+ days	"Late dinners dropped to {n} days."	S P	3
IH4	Study room	H11 up by the clear-rise	"More late-night study this {period}; sleep is study too."	S P	2
IC1	All	C01 is 7 of 7, or 90%+ of a month	"You showed up every day."	S P	3

How a screen picks: each profile headline takes its top-priority hit; each room shows up to 2 hits; the summary sentence joins the top 3 across profiles, positive first. Wording follows the copy lint in the architecture doc (no ranking words, no clinical words, no food judgement), and every template exists in English and Hinglish before it ships.

Connections engine

Connections are cross-room patterns in one student's own data, tested on a fixed list of pairs so the engine cannot go fishing for coincidences. It runs weekly on all data since day one.

Pair	Group A day	Group B day	Outcome compared
P1 Sleep → papers	bedtime before 23:30 the night before	bedtime 23:30 or later	A02 accuracy that day
P2 Breakfast → focus	breakfast before 11	no breakfast	E10 focus that day
P3 Bedtime → mood	bedtime before 23:30	later	lightness that day
P4 Desk day → next paper	desk sitting on the topic the day before	no sitting	A02 on that topic
P5 Late room → next day	30+ min in room after 23:00	none	E09 felt okay or better, next day
P6 Subject → feeling	sessions in subject X	sessions in other subjects	share "drained"

Test. Difference in means between A and B days. Show the pattern only if: each group has 14+ days (P6: 10+ sessions), the 95% bootstrap confidence interval of the difference excludes 0, and it survives Benjamini–Hochberg correction across all pairs at $q < 0.1$. The

cut-points above are fixed for everyone and never tuned per student.

Wording. "On days after you slept before 23:30, paper accuracy averaged {a}%. After later nights it averaged {b}%. Based on {n} days." Always "tends to" or "averaged", never "causes". A student can mark "doesn't feel true", which hides it for 30 days.

Food guard. A pattern is never shown if it would suggest that eating less, skipping a meal or fewer food groups goes with a better outcome.

What each dashboard reads, and its API

Three server functions serve the three screens on the design canvas. Each reads only `period_values`, `daily_breakdowns`, `insights`, `connections` and `cohort_period`, never raw room tables, and each runs `can()` first.

Student report · `getStudentReport({ period: 'day' | 'week' | 'month', date })`

Screen block	Built from
Summary sentence	top 3 insight hits across profiles
Academic profile	headline: top hit of IA rules · rows: A02, A03, A04, A09
Emotional profile	headline: top hit of IE rules · rows: E04, E09, E05, E01
Health profile	headline: top hit of IH rules · rows: H06, H07, H01, H02
What DHI connected	up to 3 shown `connections`
Exams panel	A02, A03, A05 · bars: A02 by subject, A07 by type · notes: IA3, IA4
DHI desk panel	A10, A11 count, A12 · bars: A11 by subject, A12 by mode · notes: IA2, IA6
Study sessions panel	E09, E10 · bars: E09 by subject, E10 by time band · notes: IE3, IE4, IE5
Modules panel	A15, A16, A17 · bars: A15 by category and subject · notes: IA5
Blog panel	E11, E12, E13 · bars: E15 by tag, E14 · notes: IE6
Mood panel	E02, E01, E08 · colour strip of the week · bars: E06, E07 · notes: IE1, IE2

Screen block	Built from
Meals panel	H01, H02, H03 · bars: H02 by group, H03 by slot · notes: IH1, IH3
Sleep and habits panel	H06, H07, H08 · bars: H08 by habit, H11 · notes: IH2, IH4

Response shape: `{ period, label, summary, profiles[3], connections[], rooms[{ id, kpis[], bars[], bars2[], notes[], strip? }] }`. Every number carries `{ value, logged_days, change_prev, change_start, trend }`.

Parent report · `getParentReport({ studentId, week })` — weekly, plus a monthly version.

- Academic: A02 (and by subject), A03, A11 by subject beside A02, A09, A15, teacher remarks, IA1, IA2.
- Wellbeing: only if `sharing_settings.parent_level = summary`: E04 word, E09, E01, IE2. Otherwise only E01.
- Health: H06, H07, H01, H02, IH1.
- Connections: those whose metrics the parent may see.
- "How you can help": up to 3 actions mapped from the student's top hits (for example IA2 → "ask which part of {subject} feels hard").

School report · `getSchoolReport({ classId, from, to })` — reads only `cohort_period`, plus the academic support list. Details in the next section.

Counselor view (no new screen in v1): layer states and care signals for booked students only, as in the architecture doc.

PDF export: the server renders the same payload with the same templates, so a PDF can never say something the screen doesn't.

School cohort calculations and privacy

The school sees class totals built from the same student metrics, plus one named list for academic support. Wellbeing and health totals are hidden whenever fewer than 10 students are behind a number.

School block	Formula	Minimum students
Median paper gain	median over students of A20	10

School block	Formula	Minimum students
Active students	students with C01 ⩾ 15 of 21 days (trial) or ⩾ 70% of days	none
Class accuracy, recall, understanding	Σnum ÷ Σden over all students' A02, A04; mean of A03	none
Subject table: desk time share	Σ desk minutes in subject ÷ Σ all desk minutes, whole class	none
Subject table: accuracy and change	class A02 per subject, week 3 vs week 1	none
Subject table: felt okay or better	Σ okay-or-better sessions ÷ Σ sessions per subject (E09)	10
Subject table: main reason for wrong answers	most common A07 type per subject	none
Topic heat map	class A02 per topic per week; cell colour: under 50% peach, 50–64% gold, 65%+ mint	5 students attempted
Stress by weekday	Σ stressed or anxious check-ins ÷ Σ check-ins, per weekday	10 per weekday
Health profile	median H06; mean H01 days; students with H11 > 0	10
Counselor sessions held	count	shown as "fewer than 10" below 10

Academic support list. A student is listed, A to Z, when for 2 weeks in a row A01 or A02 was Dipping, or A05 fell by 20+ points, or no room had a log for 5+ days. Each name shows its reason. Never ranked, never scored.

Wellbeing names never appear on the school screen. The counselor gets them through the care rules.

Teacher actions come from cohort rules:

- CR1: a topic under 50% with less than +5 pts over 3 weeks → "re-teach {topic}"
- CR2: one weekday's stress share ⩾ 1.6× the class mean → "look at what happens on {day}"
- CR3: the lowest subject has the smallest desk-time share → "ask students to give {subject} one more sitting"

- CR4: P2 holds for 50%+ of students with a shown result → "talk about breakfast at the parent meeting"

When things are computed

Work is done as early as it can be, so opening a report is a read, not a calculation.

When	What runs	Cost
On every room write	append to `events`; upsert that student's `daily_metrics` and `daily_breakdowns` for that day; run care rules on mood, blog and room posts	milliseconds, same request
On report open	read frozen `period_values`; compute the current, unfinished period live from `daily_metrics`; cache 5 minutes	under 300 ms target
Nightly 04:00 IST	close yesterday; freeze finished weeks (Monday) and months (the 1st); run insight rules for each student's current day, week and month; cohort rollups; academic support list	minutes for thousands of students
Sunday 04:00	connections engine on all history; weekly parent report built and sent	minutes
On formula change	rebuild the affected metric's `daily_metrics` from `events` for all students	one-off job

All jobs are idempotent per student per day, so a rerun never double-counts.

Testing and quality checks

A wrong number on a parent's or principal's report costs more trust than any feature earns, so every number is tested before it ships.

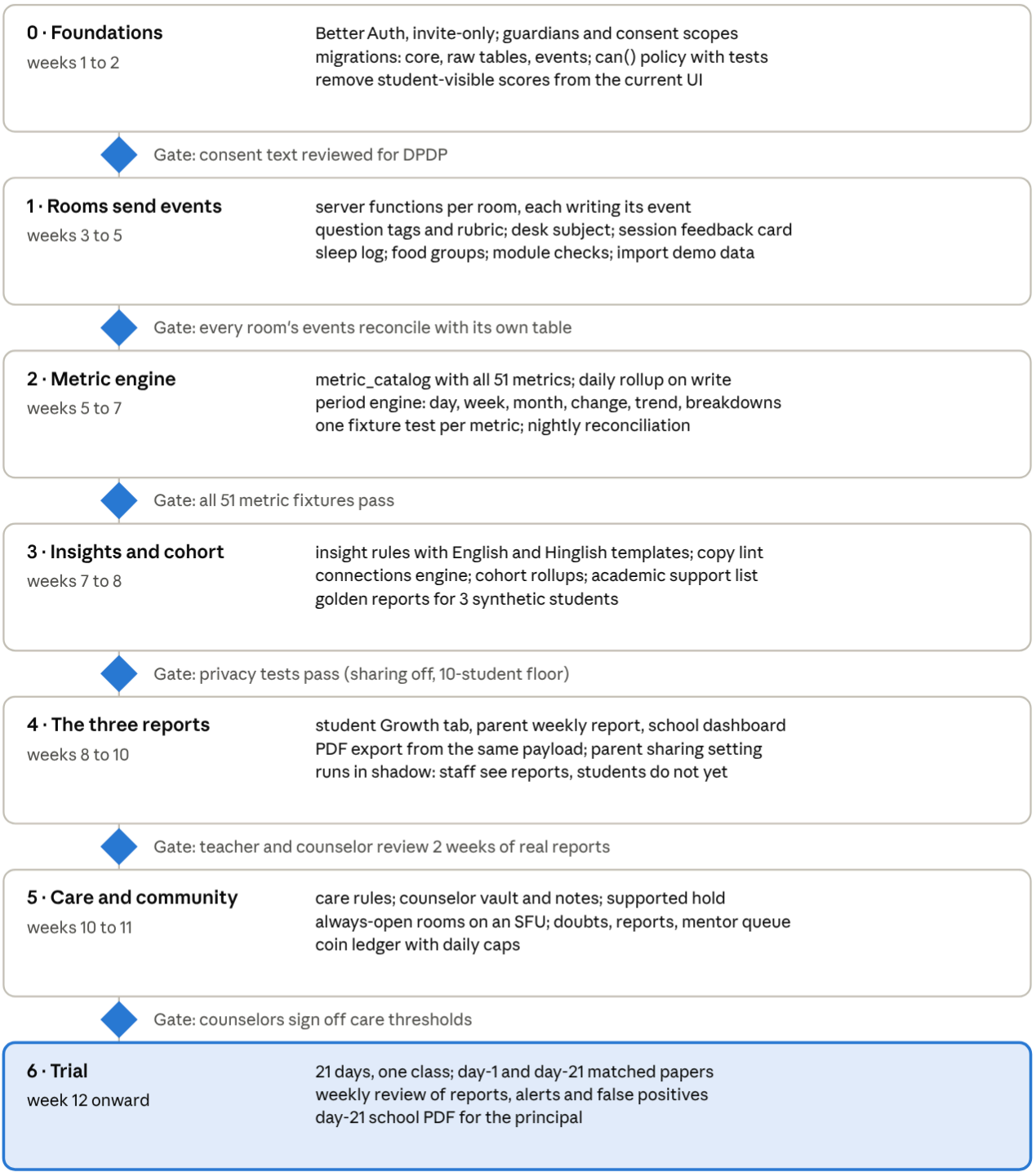
1. **One fixture per metric.** A small hand-made week of events for one student, with the expected value, change and trend written beside it. Run in CI for all 51 metrics.
2. **Golden reports.** Three synthetic students (steady, improving, struggling) with 60 days of events. Their full student, parent and school payloads are saved; any change to the output fails CI until reviewed.
3. **Insight rule tests.** For each rule, one case that fires and one that only just doesn't.

4. **Privacy tests.** The parent payload never contains mood entries or session detail when sharing is off. The school payload never contains a wellbeing number built from fewer than 10 students, and never names a student outside the support list.
5. **Copy lint.** Every template, in every language, passes the blocklist and the no-digits-except-counts rule.
6. **Reconciliation.** Nightly check that each metric's `daily_metrics` total matches a direct count from `events`; any gap pages the team.
7. **Staff review.** Before students see anything, a teacher and a counselor read the first 2 weeks of real reports for 10 students and flag anything wrong or hurtful.

Build roadmap from scratch

The order is identity first, then data capture, then the engine, then the words, then the screens; each phase ends at a gate that must pass before the next one counts. Week counts assume one full-time developer plus someone tagging questions, and are estimates.

Seven phases, about 12 weeks, from today's demo repo to the trial



build roadmap · 7 phases, 6 gates

Trial fast track (if the 21-day batch starts within 6 weeks). Build only what a 21-day trial can show:

- Phase 0 in full (login, consent and tenancy cannot be skipped).
- Phase 1 for Exams, DHI desk, Mood, Meals, Sleep and Habits, plus the session feedback card. Blog, modules and study room keep working but are not measured yet.

- Phase 2 for the 37 metrics those rooms feed (A01–A14, A20, E01–E10, H01–H10, C01–C02).
- Insight rules IA1–IA4, IE1, IE3, IH1, IH2, plus the academic support list.
- Student weekly report and the day-21 school PDF. Parents get the day-7, 14 and 21 reports as PDFs.
- Care rules and counselor alerts in full. These are never cut.

Left out on purpose: the connections engine needs 14+ days in each group, so it rarely has enough data inside 21 days anyway. Weekday stress needs 10+ students per weekday and may not show for one class.

Open decisions

Five answers change what gets built; everything else above has a default.

- [ ] **Parent sharing default:** should the emotional summary be off until the student turns it on (recommended), or on until they turn it off?
- [ ] **Trial start date:** when does the 21-day batch begin? It decides whether the fast track below is needed.
- [ ] **Who tags questions:** DHI's own team, or the school's teachers? Topic and misread tags on every question are the biggest content task.
- [ ] **Calories:** confirm "teach, don't track" for under-18s (metrics doc), since no health metric above uses amounts.
- [ ] **Counseling record:** counselor-written note or short transcript (architecture doc).